

Tribhuvan University
Faculty of Humanities & Social Sciences
OFFICE OF THE DEAN

2024

Bachelor in Computer Applications

Course Title: **Numerical Methods**

Code No: **CACS252** Semester: **IV**

Full Marks: **60** Pass Marks: **24** Time: **3 hours**

Candidates are required to answer the questions in their own words as far as practicable.

Note: Group A (Multiple Choice Questions) is not available from the source. The paper begins with Group B.

Group B

2. Explain absolute and relative error. Find the relative error of number 5.6 if both of its digits are correct.
3. On what type of equations Newton's methods can be applicable. Justify.
4. Solve the following equations by using Gauss-Jordan method. $2x+3y+4z=5$ $3x+4y+5z=6$ $4x+5y+6z=7$
5. Use the Romberg method to get an improved estimate of the integral from $x = 1.8$ to $x = 3.4$ from the data in the table with $h = 0.4$.
X : 1.6 1.8 2.0 2.2 2.4 2.6 2.8 3.0 3.2 3.4
Y : 1.61 1.82 2.02 2.22 2.42 2.62 2.83 3.03 3.23 3.43
329.96 36.59 44.70 309 534 563 481
6. Write a program to compute $\int_0^{\pi/2} \sin x \, dx$ using Simpson's 1/3 rule.
7. Using Runge-Kutta method of 4th order solve the following equation taking each step $h = 0.1$ $\frac{dy}{dx} = 4xy - xy$ given $y(0) = 3.3$ calculate y at $x = 0.1$ and 0.2 .
8. Solve the laplace equation $U_{xx} + U_{yy} = 0$ for the following square mesh with the boundary values.

Group C

Attempt any TWO questions [2x10=20]

9. Solve the given set of linear equations using Dolittle LU decomposition method:
 $3x_1 + 2x_2 + x_3 = 10$
 $2x_1 + 3x_2 + 14x_3 = 14$
10. Define initial value problems and final value problems. Using heun's method, find value of y when $x=0.3$ given that $\frac{dy}{dx} = x + y$ and $y=1$ when $x=0$.
11. How can we use Laterpolation techniques (methods) to approximate the value of the integral for the functions whose antiderivative can't be found? Explain. Write a program to solve $\sin x - 2x + 1 = 0$ using Bisection method.